

Autonomous Driving: Manual Driving Will Be as Primitive as Horse Riding

Three Camps

Tesla FSD

Cumulative: **8B+ miles**
2026 first 50 days: 1B miles
Safety: 5.1M miles/serious crash (human avg 0.7M)
2026.1: Austin driverless robotaxi launch
Pure vision · End-to-end neural network · v13+
Cybercab target: 2M units/yr

Waymo

Rider-only miles: **220.6M**
Weekly miles: 4M (Jun 2026)
Operating: 11 cities · 1,400 sq mi
90% of miles in Phoenix/Bay Area/LA
Multi-sensor (LiDAR+radar+cameras)
Pure robotaxi, no consumer vehicles

China Fleet

Baidu Apollo: **23M+ rides (Jun 2026)**
Pony.ai: Full driverless permits in 4 major cities
WeRide: Guangzhou 24/7 fully driverless ops
BYD: Free ADAS standard (incl. \$9,500 Seagull)
2.3M vehicles equipped · 150M km/day data
Huawei ADS · XPeng L3 · Li Auto L3 permit

Historical Parallel: Horse → Car Transition

1900: 21.5M horses vs 4,192 cars → 1912: NYC cars outnumber horses → 1920: 13,800 carriage makers down to 90 (**99.3% eliminated**)
Lost jobs: coachmen, farriers, harness makers, feed dealers · Key disruption period just 10 years (1907-1917) · Often cited as tech replacement causing regional economic shocks

Social Impact

Jobs

Global: **~100M** driving-related jobs
China: ~40M · US: ~5M
EU: ~6M · India: ~15M
Goldman Sachs: US full adoption → 25k jobs lost/month

Urban Space

Cars parked 95% of the time
1 shared AV ≈ replaces 11 traditional cars
14% of LA land is parking lots

Safety

93% of accidents caused by human error
US annual traffic deaths: ~40,000
AVs could theoretically eliminate most

Access

Elderly/disabled independent travel
+2M job opportunities
\$19B/yr healthcare savings

Timeline

2026

Geofenced robotaxis
Tesla · Waymo · Baidu

2030

20% new cars with L4
Dozens of cities operating

2035

57% new cars with advanced AV
China: 1.9M robotaxis

2045-2060

L5 full autonomy mainstream
Manual driving = horse riding

The carriage industry was 99.3% eliminated in 10 years. The disruptive decade for driving may be now through 2035.

BYD puts advanced ADAS free in a \$9,500 car — when tech isn't a luxury, adoption outpaces all forecasts.